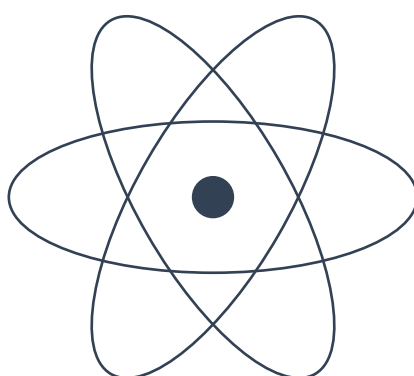


SPhOH

SAUDI PHYSICS OLYMPIAD HUB



Algebra I

Progressive Problem Set (40 Problems)

Olympiad Introductory Training • Math

SPhOH Team

FIRST EDITION • 2026

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About This Problem Set

This problem set is part of the foundational training resources of the **Saudi Physics Olympiad Hub (SPhOH)**.

It contains 40 structured problems arranged strictly by increasing difficulty, from single-step foundational algebra to challenging multi-step physics derivations.

Difficulty Levels

- **Level I** — Direct evaluations, single-step variable isolations, and standard formulas.
- **Level II** — Multi-step algebraic manipulations, systems of equations, and exponent rules.
- **Level III** — Coupled non-linear equations, approximations, and multi-variable physics derivations.

Algebra I

Topic Note. Mastering basic algebra is essential for physics. Problems are ordered sequentially from fundamental arithmetic operations to high-level system derivations and physical approximations.

1.1 Pure Algebraic Manipulations

PROBLEM 1

LEVEL I

Simple Linear Equation

Solve for x :

$$3x - 7 = 11$$

PROBLEM 2

LEVEL I

Pure Quadratic Equation

Solve for x :

$$2x^2 - 18 = 0$$

PROBLEM 3

LEVEL I

Simple Radical Equation

Solve for x :

$$\sqrt{x + 4} = 3$$

PROBLEM 4**LEVEL I****Simplifying Algebraic Fractions**

Simplify the fraction completely, assuming $x \neq 2$:

$$\frac{x^2 - 4}{x - 2}$$

PROBLEM 5**LEVEL I****Factoring Common Terms**

Factor completely:

$$3x^3 - 12x$$

PROBLEM 6**LEVEL I****Difference of Squares Mental Trick**

Calculate the value of $105^2 - 95^2$ without using a calculator.

PROBLEM 7**LEVEL I****Linear Inequality**

Solve for x and state the solution set:

$$-2x + 5 > 1$$

PROBLEM 8**LEVEL II****Factoring Quadratic Trinomial**

Solve for all real values of x :

$$x^2 - 5x + 6 = 0$$

PROBLEM 9**LEVEL II****System of Linear Equations**

Solve the system for x and y :

$$\begin{aligned}x + y &= 10 \\ 2x - y &= 5\end{aligned}$$

PROBLEM 10**LEVEL II****Variable Substitution**

Given $u = x - 1$, substitute u into the equation and solve for x :

$$(x - 1)^2 - 4 = 0$$

PROBLEM 11**LEVEL II****Exponents and Simplification**

Simplify the expression into the form 2^k :

$$\frac{2^5 \cdot 4^2}{8}$$

PROBLEM 12**LEVEL II****Proportionality Relations**

If y is directly proportional to x^2 and $y = 12$ when $x = 2$, find y when $x = 4$.

PROBLEM 13**LEVEL II****Clearing Fractional Denominators**

Solve for x :

$$\frac{1}{x} + \frac{1}{2x} = 1$$

PROBLEM 14**LEVEL II****Logarithmic Equation**

Solve for x :

$$\log_{10}(x) + \log_{10}(2) = 1$$

PROBLEM 15**LEVEL III****Completing the Square**

Rewrite $x^2 + 6x + 2 = 0$ in the form $(x + h)^2 = k$ and solve for x .

PROBLEM 16**LEVEL III****Exponential Quadratic Form**

Solve for x :

$$2^{2x} - 5 \cdot 2^x + 4 = 0$$

PROBLEM 17**LEVEL III****Nested Radical Equation**

Solve for all real x :

$$\sqrt{x+3} = x-3$$

PROBLEM 18

LEVEL III

Non-linear System of Equations

Solve the following system for x and y assuming $x, y > 0$:

$$\begin{aligned}xy &= 12 \\ x^2 + y^2 &= 25\end{aligned}$$

PROBLEM 19

LEVEL III

Rational Expression Decomposition

Express the fraction as a sum of partial fractions:

$$\frac{5x-1}{x^2-1}$$

PROBLEM 20

LEVEL III

Logarithmic System

Solve for x and y :

$$\begin{aligned}\log_2(x) + \log_2(y) &= 3 \\ x - y &= 2\end{aligned}$$

1.2 Physical Algebra Applications

PROBLEM 21

LEVEL I

Constant Speed Relation

An object travels distance $d = vt$. If speed v is doubled while distance d stays constant, what happens to time t ?

PROBLEM 22

LEVEL I

Ideal Gas Temperature Isolation

The ideal gas law is $PV = nRT$. Isolate T in terms of P , V , n , and R .

PROBLEM 23

LEVEL I

Work Done by a Constant Force

Work is $W = Fd \cos(\theta)$. Find W if $F = 50 \text{ N}$, $d = 4 \text{ m}$, and $\theta = 60^\circ$ ($\cos(60^\circ) = 0.5$).

PROBLEM 24

LEVEL I

Thermal Heat Capacity

Heat energy is $Q = mc\Delta T$. Isolate ΔT in terms of Q , m , and c .

PROBLEM 25

LEVEL I

Hydrostatic Pressure Depth

Pressure at depth h is $P = P_0 + \rho gh$. Isolate depth h in terms of P , P_0 , ρ , and g .

PROBLEM 26

LEVEL I

Photon Energy Formula

Photon energy is $E = hf$. Substituting speed of light $c = f\lambda$, write E strictly in terms of h , c , and λ .

PROBLEM 27

LEVEL I

Rotational Energy Velocity

Rotational kinetic energy is $K_{\text{rot}} = \frac{1}{2}I\omega^2$. Isolate ω in terms of K_{rot} and I , assuming $I > 0$.

PROBLEM 28

LEVEL II

Kinematic Time Isolation

A car accelerates from rest ($v_0 = 0$) such that $d = \frac{1}{2}at^2$. Isolate time t in terms of d and a .

PROBLEM 29

LEVEL II

Ohm's Law Power Formula

Given $V = IR$ and $P = VI$, eliminate V to write power P in terms of I and R .

PROBLEM 30

LEVEL II

Resistors in Parallel

Two resistors R_1 and R_2 in parallel satisfy:

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2}$$

Solve explicitly for R_{eq} .

PROBLEM 31

LEVEL II

Gravitational Inverse-Square Law

Newton's law of gravitation is $F = G \frac{m_1 m_2}{r^2}$. If distance r is doubled, by what factor does force F change?

PROBLEM 32

LEVEL II

Conservation of Energy Speed

Equate potential energy and kinetic energy: $mgh = \frac{1}{2}mv^2$. Solve for v in terms of g and h .

PROBLEM 33

LEVEL II

Kinematic Distance Formula

Given $v^2 = v_0^2 + 2ad$, solve for displacement d in terms of v , v_0 , and a .

PROBLEM 34

LEVEL II

Spring Potential Energy Stretch

Spring energy is $U_s = \frac{1}{2}kx^2$. Isolate stretch x in terms of U_s and k , assuming $x > 0$.

PROBLEM 35

LEVEL III

Thin Lens Equation Isolation

The thin lens formula is:

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

Isolate image distance d_i in terms of f and d_o .

PROBLEM 36**LEVEL III****Centripetal Acceleration and Frequency**

Centripetal acceleration is $a = \frac{v^2}{r}$ and speed is $v = 2\pi r f$. Substitute v into a to express acceleration in terms of π , r , and f .

PROBLEM 37**LEVEL III****Atwood Machine Acceleration**

Solve the force system for acceleration a :

$$T - m_1 g = m_1 a$$

$$m_2 g - T = m_2 a$$

PROBLEM 38**LEVEL III****Pendulum Period Scaling**

The period of a pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$. If length L quadruples ($L' = 4L$), by what factor does T change?

PROBLEM 39**LEVEL III****Simple Binomial Approximation**

For small $x \ll 1$, $(1 + x)^n \approx 1 + nx$. Use this to approximate $\frac{1}{1+x}$ when x is small.

PROBLEM 40**LEVEL III****Relativistic Kinetic Energy Limit**

The relativistic energy equation is $E = \gamma mc^2$, where $\gamma = \frac{1}{\sqrt{1-v^2/c^2}}$. Express γ in terms of v and c , then use the binomial approximation $(1 - y)^{-1/2} \approx 1 + \frac{1}{2}y$ (for $y = v^2/c^2 \ll 1$) to show that kinetic energy $K = (E - mc^2)$ reduces to classical kinetic energy $\frac{1}{2}mv^2$.

Hints

@hintlist